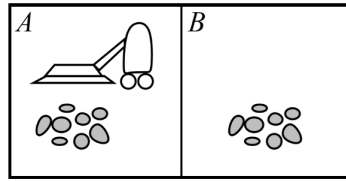


1. State-based Search Formulation (1.5 pts)

Consider the scenario where a vacuum cleaner agent may sense two locations A, B and clean up the dirt in each location it is in.

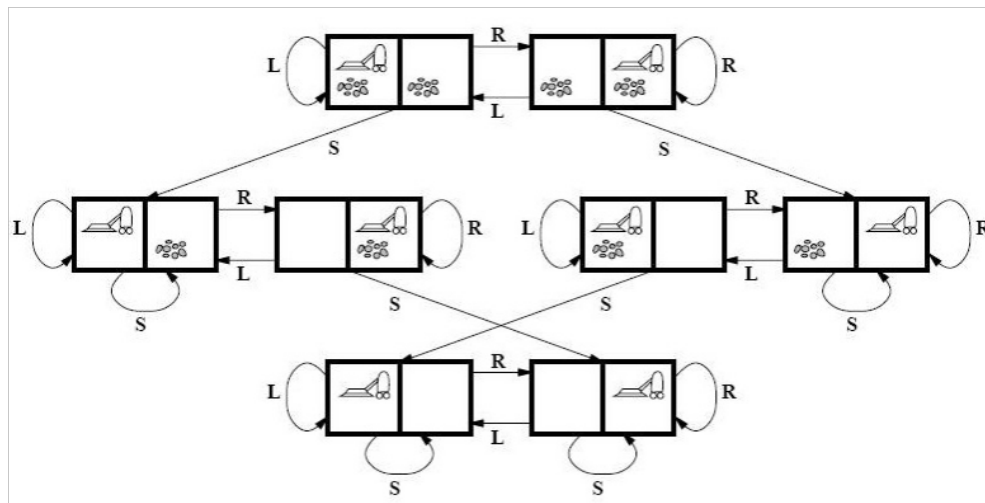


The agent can perform three actions:

- **Left (L)**: move the cleaner to the left, stay if the movement is not possible.
- **Right (R)**: move the cleaner to the right, stay if the movement is not possible.
- **Suck (S)**: clean up the dirt of the current location if it is dirty.

Consider this situation as a state-based model. **Draw the transition graph.**

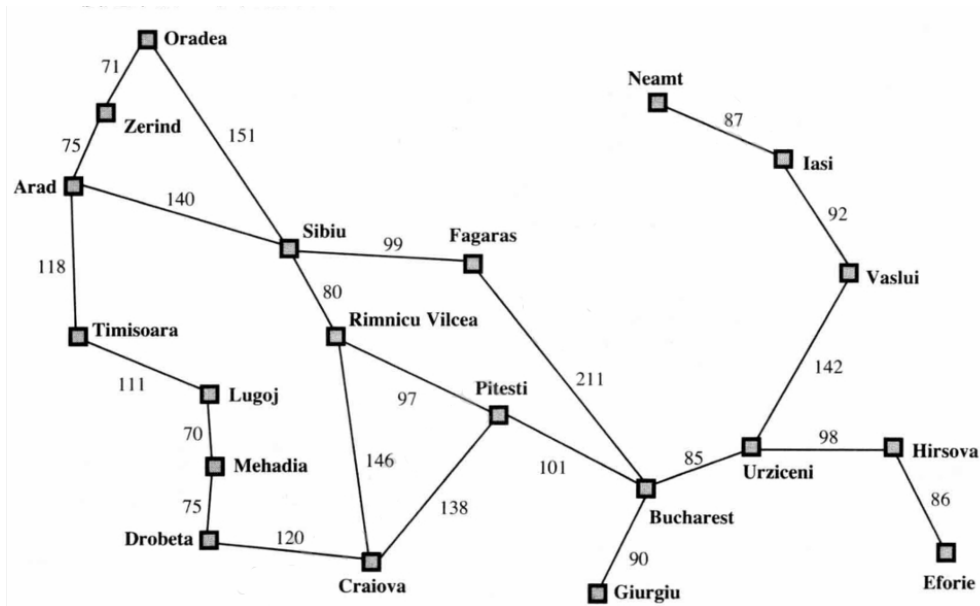
Solution



2. A* Search (2 pts)

Apply A* search algorithm to find a path from Arad to Bucharest. We use the straight-line distance from a node to Bucharest as the heuristic (listed below). **Report the first five paths removed from the frontier.**

Location	$h(v)$	Location	$h(v)$	Location	$h(v)$	Location	$h(v)$
Arad	366	Eforie	161	Rimnicu Vilcea	193	Hirsova	151
Urziceni	80	Iasi	226	Vaslui	199	Lugoj	244
Zerind	374	Mehadia	241	Bucharest	0	Fagaras	176
Sibiu	253	Giurgiu	77	Timisoara	329	Neamt	234
Craiova	160	Oradea	380	Drobeta	242	Pitesti	100

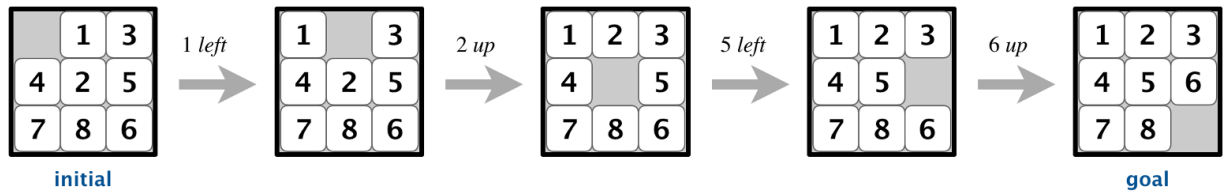


Solution

1. (Arad)
2. (Arad, Sibiu)
3. (Arad, Sibiu, Rimnicu Vilcea)
4. (Arad, Sibiu, Fagaras)
5. (Arad, Sibiu, Rimnicu Vilcea, Pitesti)

3. 8-puzzle Board Game (1.5 pts)

Consider placements of numbers 1,...,8 to a 3×3 board where each number occupies a cell, and one cell is unoccupied. At each step, the player may move one number from an adjacent cell to the unoccupied cell. The goal is to move the board from an initial state to a goal state displayed below.



Propose two admissible heuristics. Explain why they are admissible.

Solution

- $h_1(s)$: the number of numbers that are not at the final position in state s . For each number that is not at its final position, we need no less than one move to reach its final position. So, $h_1(s)$ will never overestimate the cost of the optimal solution.
- $h_2(s)$: the sum of the Manhattan distances from each cell to its final position in state s . For each number that is not at its final position, the Manhattan distance to its final position is the length of the shortest path to its final position, thus is always no greater than the optimal solution.